



Solar Dryer



Towards Sustainable Future
ZeniTEK

Organic. Hygienic. Efficient

About Us

Established in 2021 in Erode, Tamil Nadu, **Zenitek** is a pioneering force in renewable energy technology. We design and manufacture advanced systems that reduce carbon footprints and empower industries, institutions, and agriculture with clean, efficient solutions.

What we do

Category	Solutions
Solar Thermal Systems	▪ Evacuated Tube Collectors ▪ Parabolic Trough Collectors ▪ Dish Collectors & Scheffler Systems <i>Designed for high-efficiency process heat applications</i>
Agri-Solar Innovations	▪ Solar-Powered Paddy Transplanters ▪ Cold Storage Units ▪ Solar Dryers
Photovoltaic Solutions	▪ PV Panels & Systems ▪ Test Rigs for PV Testing & Certification Labs
Research & Development	<i>In collaboration with IITs and top institutions:</i> ▪ Demonstration Models ▪ Prototype Test Rigs ▪ Turnkey Commercialization Systems

Why We Entered Solar Drying Solutions

Recognizing the post-harvest challenges faced by farmers and agri-businesses, especially the heavy losses due to improper drying methods, ZeniTek ventured into solar dryers. By combining scientific drying principles with renewable energy, we aim to offer reliable, cost-effective, and eco-friendly drying solutions that enhance product quality, shelf life, and market value—while supporting sustainable agriculture.

How ZeniTek Differs from Others

While many solar dryer manufacturers offer generic models, ZeniTek stands apart by engineering customized, data-driven systems tailored to the specific drying needs of each crop and climate condition. We integrate temperature and humidity control, automated airflow management, and smart energy optimization to ensure consistent drying performance. Our approach combines scientific research, field testing, and user feedback, making our dryers more reliable, scalable, and farmer-friendly. Beyond just selling equipment, we partner with users to deliver long-term performance, training, and after-sales support.

How Our Solar Dryer Works

ZeniTek’s solar dryer uses the greenhouse effect and controlled airflow to remove moisture efficiently from agricultural produce. Sunlight enters through the transparent or translucent roof, heating the air inside the drying chamber. This warm air absorbs moisture from the products and is continuously circulated using solar-powered or electric fans to ensure uniform drying.

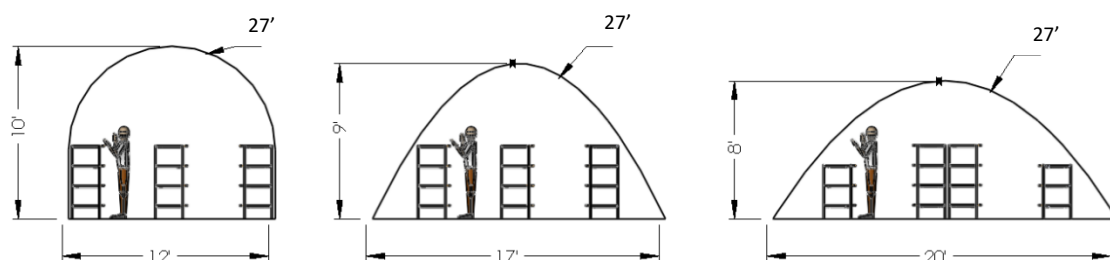
During daytime, the system maintains optimum drying temperatures using natural solar heat. At night or during cloudy weather, the design ensures sufficient insulation and passive airflow control to maintain appropriate conditions and prevent moisture buildup. Sensors monitor temperature, humidity, and airflow, while the smart control system adjusts operations for consistent drying performance and energy efficiency.

Understanding Solar Dryer area requirement

The area required for a solar dryer depends on several factors, including the type of crop, its initial and final moisture content, drying time, dryer shape and daily solar availability.

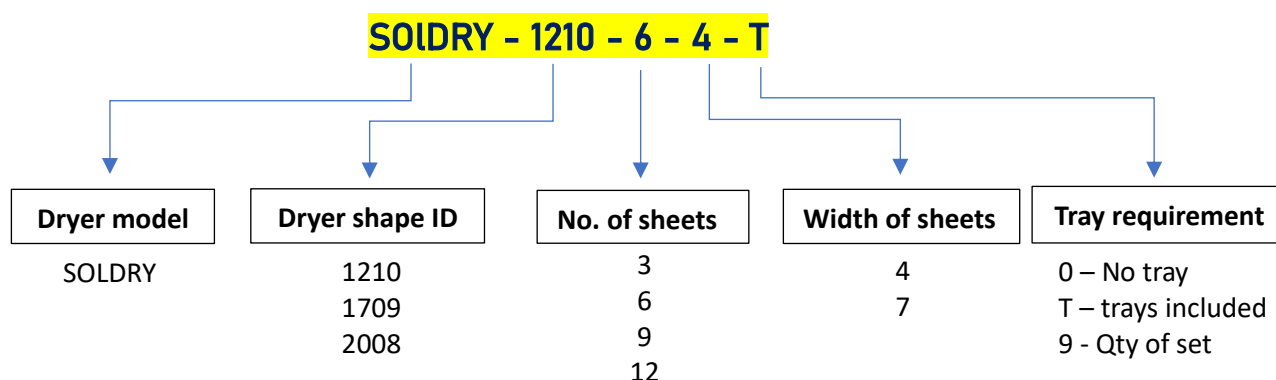
As a general guideline:

- 1 to 3 square feet of drying area is required per kilogram of fresh produce.
- Leafy greens** and thin-sliced fruits require more surface area (**300g / 1 Sqft**) and faster drying, while denser produce like **coconut or turmeric (1 – 1.5 kg/ 1 Sqft)** can be stacked more efficiently.
- Batch size, loading density, and number of drying trays also influence total dryer size.
- Width of the sheet: **4 ft for high wind area, 7 ft for low wind area** (steel structural cost reduced).
- Shape of dryer as below (**For same cost and material**)

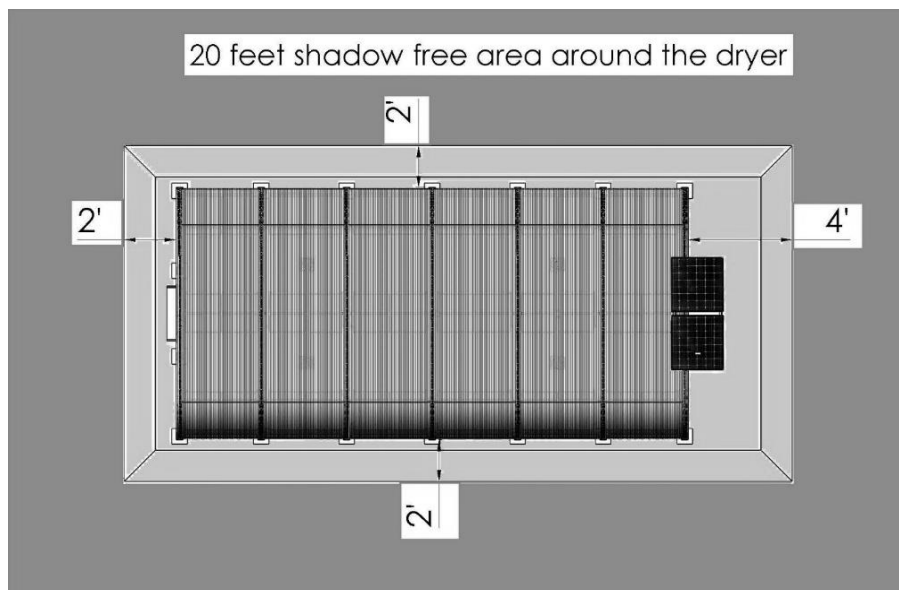


Dryer Shape		D		Narrow parabola		Widened parabola	
SOLDRY Model		1210		1709		2008	
Suitable for		Tray drying		Tray and floor drying		Floor drying	
Accessibility		Easy		Moderate		Hard in corners	
Sheet width in ft		4	7	4	7	4	7
Area per sheet in Ft ²		48	84	68	119	80	140
No. of sheets	X3	144	252	204	357	240	420
	X6	288	504	408	714	480	840
	X9	432	756	612	1071	720	1260
	X12	576	1008	816	1428	960	1680
	No. of tray Stands (No. of trays per stand) (6.25 Sqft/tray)						
	X3	9 (4)	15(4)	9 (4)	15(4)	6 (3) ; 6(4)	10 (3) ; 10(4)
	X6	18 (4)	30(4)	18 (4)	30(4)	12 (3) ; 12(4)	20 (3) ; 20(4)
	X9	27 (4)	45(4)	27 (4)	45(4)	18 (3) ; 18(4)	30 (3) ; 30(4)
	X12	36 (4)	63 (4)	36 (4)	63 (4)	24 (3) ; 24(4)	42 (3) ; 42(4)

Ordering code



Area required around solar dryer

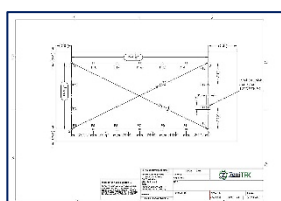


Roof top



Ground

How we install the dryer



Foundation drawing
Provide by ZeniTEK



Side wall built and
soil filled



Concrete filled with
holes for post



Post aligned leveled
and concrete filled

All Civil related work - Buyer scope



Installation Finished



PV panel fixed and
wiring completed



Polycarbonate sheet
beading fixed



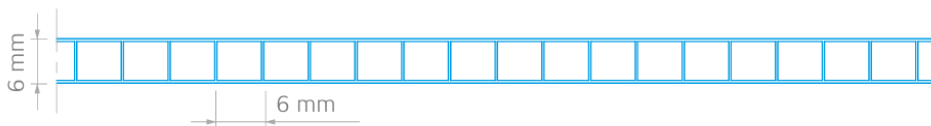
GI frame
assembled

What customer need to do while installing dryer

- ✚ Provide local guidance to installation team for transportation, material purchase if any.
- ✚ Provide single phase electrical power supply up to site temporarily or permanent if required.
- ✚ Mason support to fix the foundation post.

Know about cover material - Polycarbonate sheet

Polycarbonate sheet features a unique 2 side proprietary surface treatment designed to protect the Sheet against the degrading effects of ultra-violet radiation in natural sunlight. Available in 6 mm thickness, easy to cut and install sheets are treated with proprietary UV protection on both sides to help ensure long-term optical quality and high light transmission against outdoor weathering and minimize installation mistake costs.



Impact Strength

Polycarbonate sheet has outstanding impact performance over a wide temperature range, -40°C to $+120^{\circ}\text{C}$, and also after prolonged outdoor exposure.

Hail Resistance

As a roof glazing material Polycarbonate sheet provides outstanding impact performance even in severe weather such as windstorm, hail-stones, snowstorms and ice formation. Virtually unbreakable Polycarbonate sheet is able withstand harsh weathering conditions as well as to the subsequent temperature change to sunny conditions without breaking or buckling

Light Transmission

The wavelength of the sun affecting a thermoclear sheet material range from 280 up to 2140 nanometers. This optical window is divided into the following sections:

UV-B Middle ultraviolet region 280 – 315 nm

UV-A Near ultraviolet region 315 – 380 nm

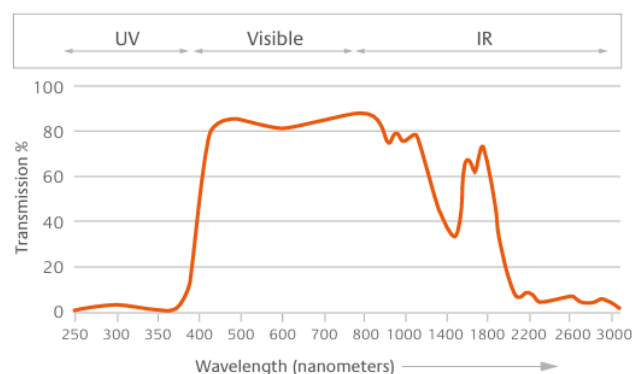
Visible light region 380 – 780 nm

Near infra-red region 780 – 1400 nm

Middle infra-red region 1400 – 3000 nm

As shown in Graph, Polycarbonate sheet has the highest transmission in the visible light. Despite transmitting visible light very well, Polycarbonate sheet is almost opaque to radiation in the UV and far infra-red region. This useful shielding property can prevent discolouration of sensitive materials such as fabrics or other organic materials placed under or behind Polycarbonate sheet glazing

Light Transmission Spectrum of Polycarbonate sheet

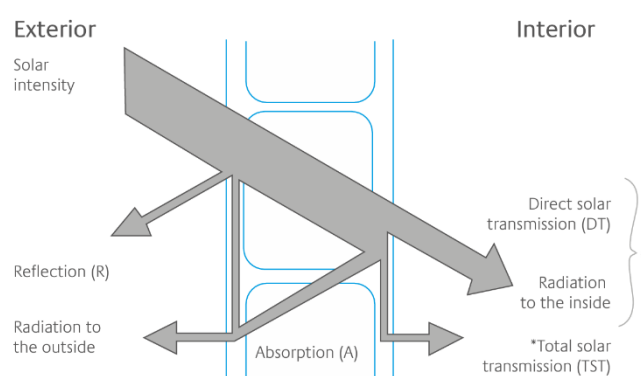


Temperature Increase

Inside the Building Sunlight entering the building heats the air both directly and through absorption by the framework, furniture, etc., and is released as infra-red energy. In combination with the insulating properties of Polycarbonate sheet, this prevents heat escaping faster than it is created causing a temperature increase – the so-called 'greenhouse effect'. The temperature can be controlled by venting hot air with exhaust fans.

Solar Heat Gain

The solar radiation reaching the sheet is reflected, absorbed and transmitted, as shown in image. The greatest proportion is transmitted and the total solar transmission (TST) is the sum of the direct transmission (DT) and the inwardly released part of the absorbed energy (A).



*All information's provided in this page is taken from technical manual of LEXAN

Understanding Dryer Controllers

Dryer controllers are essential components that automate and regulate the drying environment inside a solar dryer. They help maintain optimal temperature, humidity, and airflow, improving drying efficiency and product quality while reducing the need for manual intervention. Depending on the level of automation and data handling, controllers can range from basic timers to advanced smart systems.

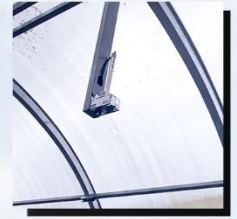
The table below compares different types of dryer controllers offered by ZeniTek across key features and functionalities.

Controller model	2T	2TRH	AUTO
Controllers	Analog timer, Temperature controller	Analog timer, Temperature and humidity controller	PLC with T+Rh module and HMI
Suitable for	Household dryer	Household and small scale Tunnel dryer	Tunnel dryer
24v DC solar power	✓	✗	✓
220V AC Grid	✗	✓	✓ optional
Battery Backup	✓	✗	✓
Timer control	✓	✓	✓
Temperature control	✓	✓	✓
Humidity Control	✗	✓	✓
Rain detection	✗	✗	✓
Display	Inbuilt in Controller	Inbuilt in Controller	HMI touch screen
Control Program	Fixed	Fixed	Programmable
Real time clock	✗	✗	✓
Trend graph (12 hr)	✗	✗	✓
Datalogging	✗	✗	✓
Heater integration	✗	✗	✓
Dehumidifier integration	✗	✗	✓
Day night detection	✗	✗	✓
Ambient Rh based control	✗	✗	✓
Image			

Features of Zenitek's Commercial Solar dryer

Aluminium beading with EPDM rubber seal

(Top and bottom aluminium beading with rubber seal holds the sheet firmly and arrest air leakage and rain water leakage)



Indoor sensor

(Measures inside air temperature and humidity)

Runs with solar PV energy

(Emergency grid support provided)

Outdoor sensor

(Measures outside air temperature and humidity)



Exhaust fan unit

(8" DC fans with rain water arrester completely made of GI sheet)



Control panel

(PLC & HMI integrated control with multiple mode of operation and data logging)

Air intake vent

(With manual shutter to control air flow and rain cover)

Double walled polycarbonate sheet

(2UV solar grade with better thermal insulation and impact proof covering sheet)

GI steel structure bolted assembly

(Pre fabricated GI frames assembled with bolt and nut. Rust free and long life.)

SS 304 grade perforated sheet tray

(3 or 4 tray per GI stand . Center row stand provided with trolley wheels for easy movement)



6" DC fan for Internal air circulation



ZeniTek Household Solar Dryer – Compact, Clean, and Convenient

The ZeniTek Household Solar Dryer is a compact and easy-to-use solution for drying small quantities of fruits, vegetables, herbs, and spices at home. It uses solar energy to dry food naturally, preserving nutrients and flavor without chemicals.

◆ Key Features:

- Runs on sunlight, no electricity needed
- Portable and lightweight design
- UV-protected transparent cover for better heat retention
- Ventilation system for uniform drying
- Ideal for daily kitchen use, balcony, or backyard

Perfect for home gardeners, homemakers, or anyone who wants to dry food hygienically and economically.



Model	SUNDRIY MINI	
	COMPACT	JUMBO
Size	2ft(w) x 4ft(L) x 1ft (H)	4ft(w) x 4ft(L) x 3ft (H)
No.of trays (total Area)	1 (6.25 ft ²)	4 (25 ft ²)
Adaptable no. of trays	2 (on request and added cost)	6 or 8 (on request and added cost)
Tray material	SS 304 food grade perforated sheet	
Dryer frame	GI sheet and GP pipes welded together (Rust free)	
Cover material	6mm thick double walled polycarbonate sheet with 2 side UV protected	
Caster wheels	4 no's with break locks	
Exhaust Fan	12v Dc	24V DC
Solar Panel	10W 12V	10W 24V
Fan Controller	No (Direct run based on solar power)	Timer and temperature controller
Battery	No	24v 7Ah
Charge controller	No	Victron MPPT



Photo Gallery



Solar Dryer technical enquiry form

• Mandatory fields

S. No	Description	Details
1	Name of the person*	
2	Address*	
3	Contact number*	
4	Location (google map) & Site photo*	
5	Dry Product i.e. vegetables, fruits, others*	
6	Product wet state moisture (% in weight basis)	
7	Product dry state moisture (% in weight basis)	
8	Quantity of product/day/batch*	
9	Drying Temperature (°C)	
10	Existing drying method (cabinet, conveyor, open land area)	
11	Fossil fuel details if used	
12	Daily energy consumption (Electricity-units/liters/KJ)	
13	Existing drying duration	
14	Night time operation (Yes or No)*	
15	Available Shadow free area (E-W & N-S)*	
16	System wants to install roof top or ground floor If rooftop, please mention the height from the ground level*	
17	One kg of product accommodate area in sq.ft*	
18	For drying the product tray is required?*	
19	Comments	



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